

Clarification of the GreatWall Variants and their Target Security Levels

GreatWall is submitted with four parameter variants, namely GreatWall-128, GreatWall-192, GreatWall-256, and GreatWall-512. All four variants follow the same overall construction, including the Pylon one-way function, the VOLE-in-the-Head proof framework, and the signing and verification procedures. They differ only in their concrete parameters, which are selected to provide different security guarantees.

For the purpose of evaluation, the submitted variants are assigned to the three security levels as follows: GreatWall-192 is evaluated at Level 1, GreatWall-256 is evaluated at Level 2, and GreatWall-512 is evaluated at Level 3. GreatWall-128 is included as an additional submitted parameter variant and is not used as the representative instance for any of the three levels. And the details are summarized in Table 1.

Table 1: GreatWall variants and the corresponding target security levels.

Variant	Evaluation Level	Classical Security	Quantum Security
GreatWall-128	Additional variant	-	-
GreatWall-192	Level 1	192 bits	80 bits
GreatWall-256	Level 2	256 bits	128 bits
GreatWall-512	Level 3	512 bits	256 bits

We emphasize that the above four names identify the cryptographic parameter variants of GreatWall. They should be distinguished from the suffixes S and F used in the implementation and performance evaluation. For each GreatWall parameter variant, the S and F suffixes denote two implementation-oriented VOLEitH parameter profiles of the same underlying cryptographic construction. In particular, the S profile is optimized for shorter signatures, whereas the F profile is optimized for faster signing and verification.

For example, GreatWall-128S and GreatWall-128F are not two independent cryptographic variants. Both instantiate GreatWall-128 and use the same underlying Pylon-128 parameters; their difference lies only in the VOLEitH parameters used to obtain different trade-offs between signature size and computational performance. The same convention applies to GreatWall-192S/F, GreatWall-256S/F, and GreatWall-512S/F.

Therefore, throughout this document, GreatWall-128, GreatWall-192, GreatWall-256, and GreatWall-512 refer to the four submitted cryptographic parameter variants, while the S/F suffixes refer only to implementation profiles used for benchmarking. In particular, the variants participating in the level-based evaluation are GreatWall-192 for Level 1, GreatWall-256 for Level 2, and GreatWall-512 for Level 3.